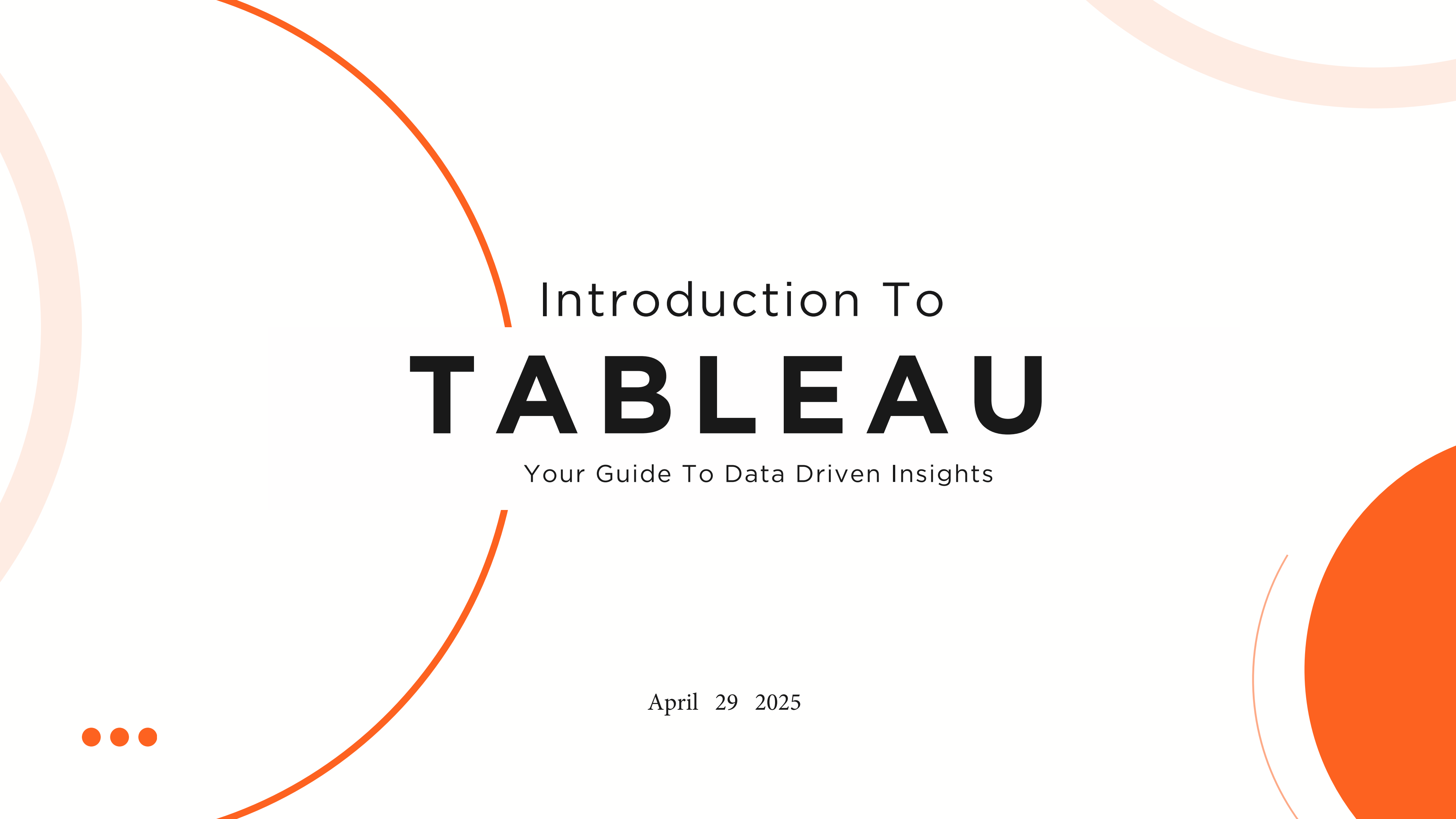




Introduction To **TABLEAU**

Your Guide To Data Driven Insights

April 29 2025





Objectives

By the end of this session, you will :

- **Understand how to connect Tableau to a MySQL table.**
 - **Display MySQL data in Tableau and create associated graphs.**
 - **Understand groupings in Tableau (similar to GROUP BY in SQL).**
 - **Execute SQL queries and visualize the results in Tableau.**
- 
- 

Connecting to MySQL Workbench

1

1. Open MySQL Workbench:

- Launch MySQL Workbench on your system.

2. Enter Username and Password:

- When prompted, enter your MySQL username and password.
- Default username: root (if not changed).
- Password: Set by you or your system administrator.(default: 1234)

3. If You Forgot Your Username or Password:

- Check Username: Navigate to the Home screen → Under "MySQL Connections", locate your connection → Click Edit Connection Settings → Check the Username field.
- Forgot Password: Follow this full guide for resetting passwords available on [MySQL's official site](#).



Connecting MySQL Workbench to an Excel File

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1. Open MySQL Workbench and go to Schemas.
2. Right-click to create a new schema (e.g., student_data) or select an existing schema .
3. Expand the chosen schema (e.g., student_data), right-click on Tables, and select **Table Data Import Wizard**.
4. In the Import Data panel, browse to the required Excel file and click Next.
5. Verify the new table name (e.g., students_info) and data view, then click Next and proceed until finished.
6. Right-click on the created schema (e.g., student_data) and click Refresh All.
7. Once the new table (e.g., students_info) is visible, run the following queries to verify the data :
 - USE student_data; (Example schema name)
 - SELECT * FROM `students\_info`; (Example table name)

Connect

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File Data Server Help



Connect

Search for Data

Tableau Server

To a File

Microsoft Excel

Text file

JSON file

Microsoft Access

PDF file

Spatial file

Statistical file

More...

To a Server

Microsoft SQL Server

Oracle

Amazon Redshift

MySQL

More...

Saved Data Sources

Sample - Superstore

World Indicators

Open



Book1-Tableau



Book1



Salesforce%20Sales%...

[Open a Workbook](#)

Quick Start

Accelerators

Jumpstart your analysis with pre-built templates
[View More](#)



Salesforce Sales Cloud - ...



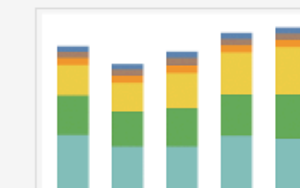
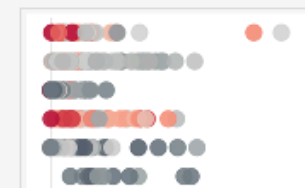
Budget Controlling



Retail Sales

Sample Workbooks

Explore what Tableau can do



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[Get Started](#)

[Tour the Tableau Environment](#)

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[Explore Now →](#)

Connecting Tableau to MySQL

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File Data Server Help

Search

Sort by Name (a-z)

Connect

Search for Data

Tableau Server

To a File

- Microsoft Excel
- Text file
- JSON file
- Microsoft Access
- PDF file
- Spatial file
- Statistical file
- More...

To a Server

- Microsoft SQL Server
- Oracle
- Amazon Redshift
- MySQL
- More...

Saved Data Sources

- Sample - Superstore
- World Indicators

Installed Connectors (71)

- Action Vector
- Alibaba AnalyticDB for MySQL
- Alibaba Data Lake Analytics
- Alibaba MaxCompute
- Amazon Athena
- Amazon Aurora for MySQL
- Amazon EMR Hadoop Hive
- Amazon Redshift
- Anaplan (deprecated)
- Apache Drill
- Azure Data Lake Storage Gen2
- Azure SQL Database
- Azure Synapse Analytics
- Box
- Cloudera Hadoop
- Databricks
- Denodo (deprecated)
- Dremio
- Dropbox
- Esri
- Exasol
- Firebird 3
- Google Analytics
- Google BigQuery
- Google BigQuery (JDBC)

MySQL

General Initial SQL

Server
localhost

Port
3306

Database
Optional

Username
root

Password
Optional

☐ Require SSL

Sign In

Additional Connectors (38)

- Google Cloud SQL
- Google Drive
- Hortonworks Hadoop Hive
- HPE Ezmeral Data Fabric (Ma
- IBM DB2
- IBM Netezza Performance Se
- Impala
- Intuit QuickBooks Online (dep
- Kyvos
- LinkedIn Sales Navigator (dep
- MariaDB
- Marketing Cloud Intelligence
- Marketo (deprecated)
- MarkLogic (deprecated)
- Microsoft SQL Server
- Microsoft SQL Server Analy
- MonetDB
- MongoDB BI Connector
- OData
- OneDrive and SharePoint Online
- Oracle
- Oracle Eloqua (deprecated)
- Oracle Essbase
- Pivotal Greenplum Database
- PostgreSQL

Other Databases (JDBC)

Other Databases (ODBC)

- Palantir Foundry by Palantir
- Qubole Hive by Qubole
- Rockset by Rockset, Inc.
- Salesforce Marketing Cloud by Tableau
- SAP SuccessFactors by Tableau
- ServiceNow by Tableau
- Sharepoint Lists (JDBC) by Tableau
- SingleStoreDB JDBC by Singlestore
- Splunk by Tableau
- SQream DB by SQream Technologies
- Starburst Enterprise by Starburst
- StarTree Tableau Connector by StarTree
- Stratio Crossdata by Stratio BD
- Yellowbrick by Yellowbrick Data

Jethro ODBC by Jethro Data

Kyligence Connector by Kyligence

MarkLogic by MarkLogic

Ocient JDBC by Ocient

OpenSearch by Amazon

Oracle NetSuite by Tableau

Connecting Tableau to MySQL

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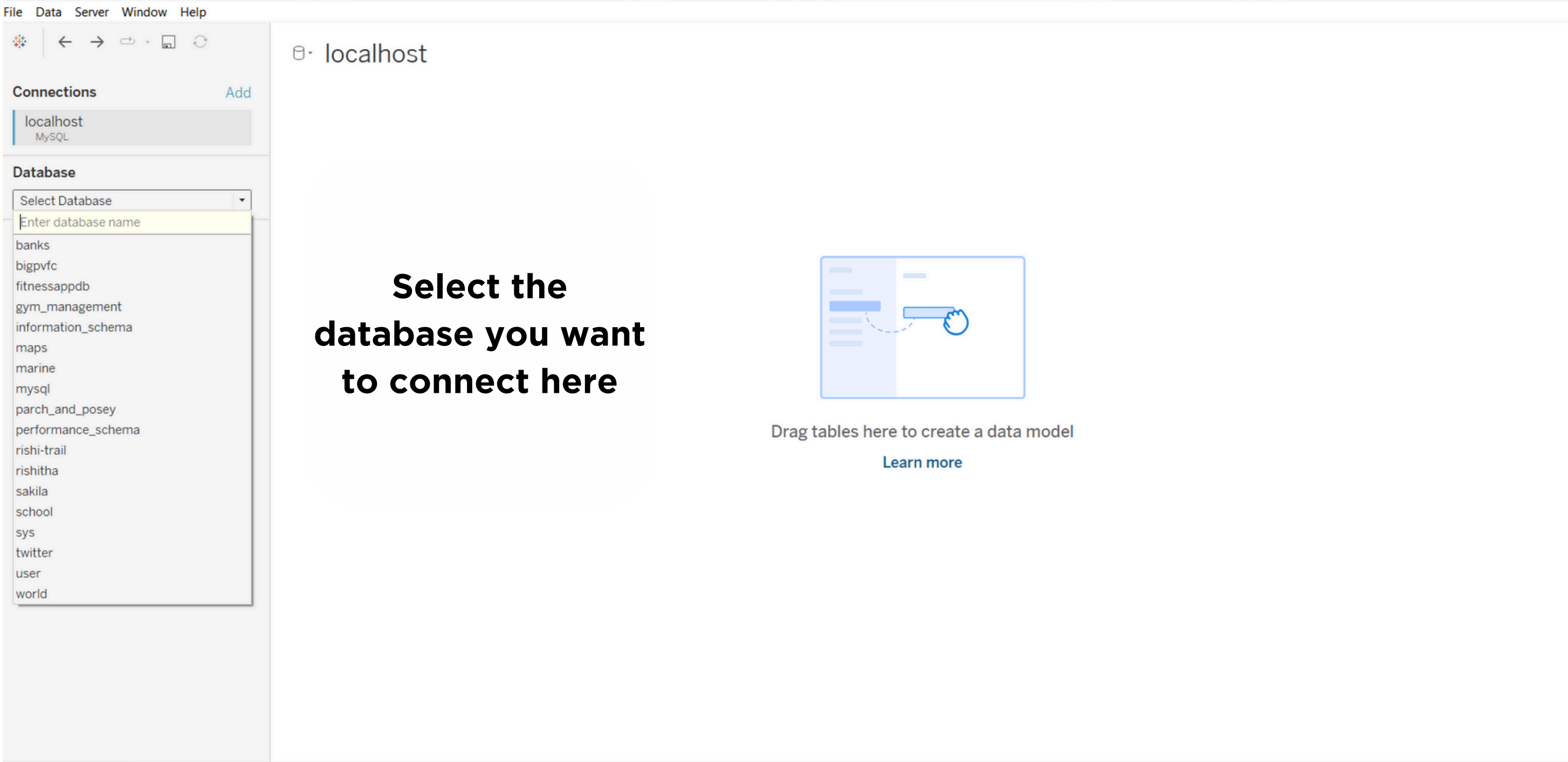
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The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane displays 'localhost MySQL'. Below it, the 'Database' pane shows a dropdown menu with 'Select Database' and a list of databases including 'banks', 'bigpvfc', 'fitnessappdb', 'gym_management', 'information_schema', 'maps', 'marine', 'mysql', 'parch_and_posey', 'performance_schema', 'rishi-trail', 'rishitha', 'sakila', 'school', 'sys', 'twitter', 'user', and 'world'. The number 5 is highlighted in a red circle next to the 'Database' pane. In the center, text reads 'Select the database you want to connect here'. On the right, a diagram shows a hand dragging a table from a list into a workspace, with text below it saying 'Drag tables here to create a data model' and a 'Learn more' link.

File Data Server Window Help

localhost MySQL

Connections Add

Database

Select Database

Enter database name

banks
bigpvfc
fitnessappdb
gym_management
information_schema
maps
marine
mysql
parch_and_posey
performance_schema
rishi-trail
rishitha
sakila
school
sys
twitter
user
world

Select the database you want to connect here

Drag tables here to create a data model

[Learn more](#)

Connecting Tableau to MySQL

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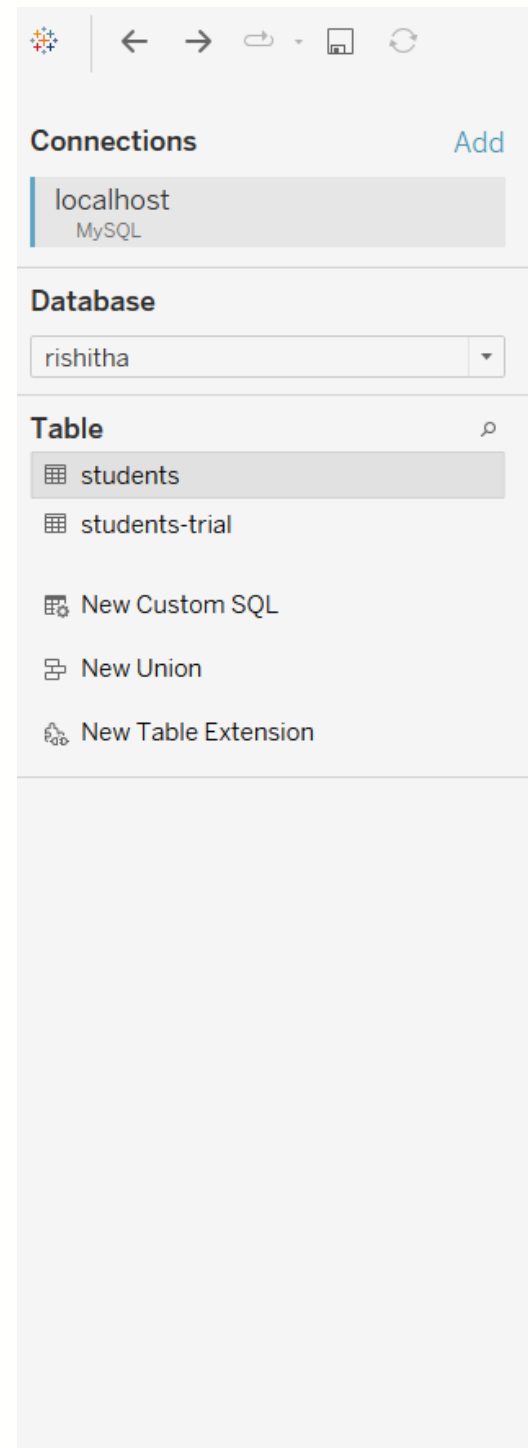
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rishitha

**Drag and drop the
desired table on to
the panel.**



Drag tables here to create a data model

[Learn more](#)

Connecting Tableau to MySQL

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localhost
MySQL

rishitha

students
students-trial
New Custom SQL
New Union
New Table Extension

students (rishitha)

students

Connection
☒ Live ☐ Extract
Filters
0 | Add

students 7 fields 200 rows 100 rows

Name
students

Fields

Type	Field Name	Physical Table	Remote Fiel...
#	Student ID	students	StudentID
Abc	Name	students	Name
#	Age	students	Age

# students Student ID	Abc students Name	# students Age	Abc students Email	Abc students Department	# students GPA	# students Graduation Ye
3336	David Palmer	19	sean43@hotmail.com	Mathematics	3.16000	
8774	Andrew Roach	23	vbecker@harvey.com	Chemistry	3.75000	
1396	Jonathan Gonzalez	22	hollydavis@gmail.com	Physics	2.95000	
6716	Kenneth Morrow	24	ganderson@wheeler-atkins.i...	Physics	3.55000	
8830	Kaitlyn Martinez	18	hayesdiane@gmail.com	Chemistry	2.29000	
5305	Tiffany Wolf	23	qanderson@taylor.com	Mathematics	3.30000	

Data Source

Sheet 1

We can see the
selected tables
data view now.

Creating graphs

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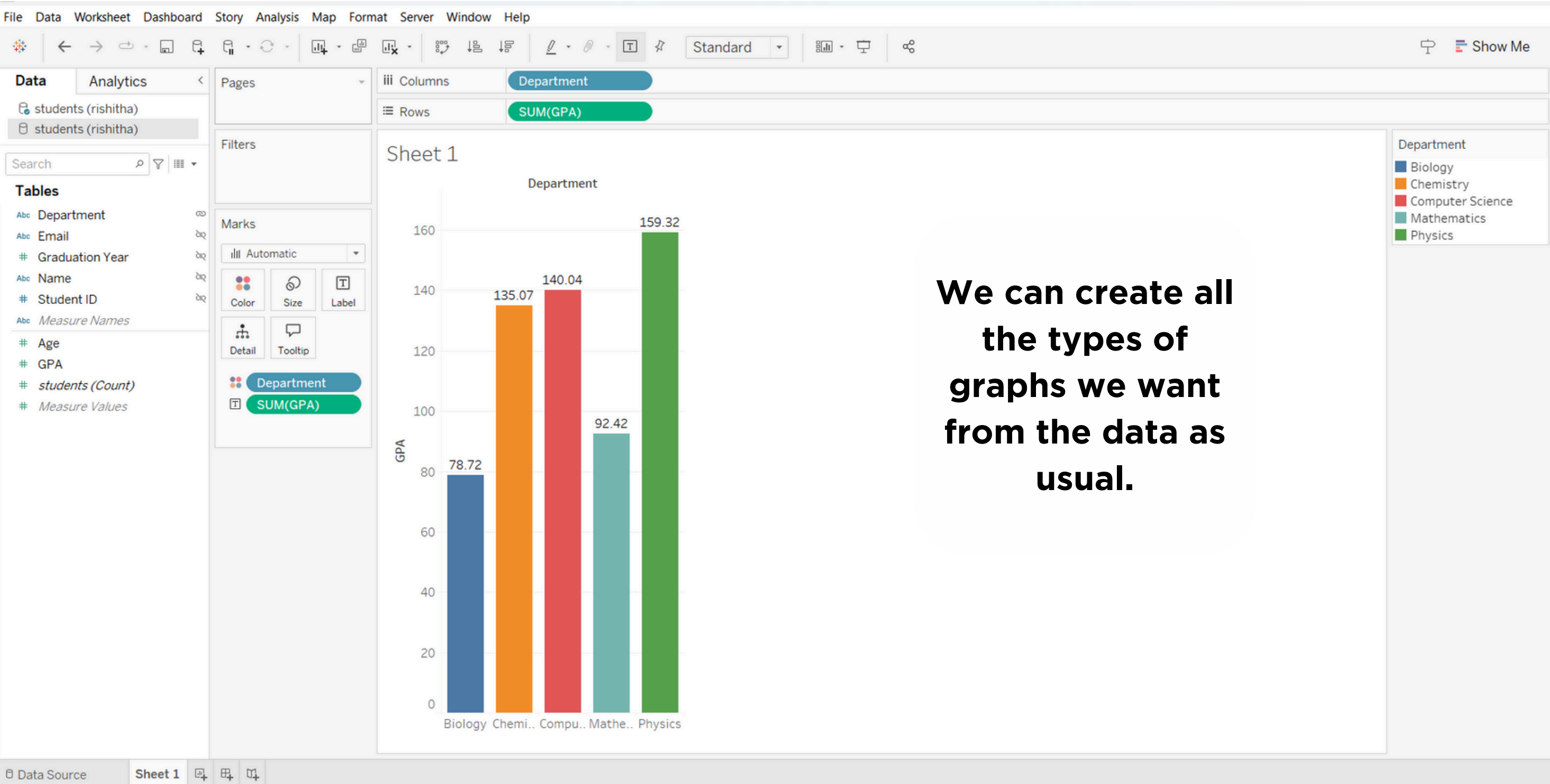
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**We can create all
the types of
graphs we want
from the data as
usual.**

Understanding Groupings in Tableau

(Similar to SQL GROUP BY)



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GROUP BY in SQL:

- Groups rows that have the same values into summary rows.
- Often used with aggregate functions (SUM, COUNT, AVG, etc.).

GROUP BY in Tableau:

- In Tableau, this is done using Dimensions and Measures.
- Example: Group data by category (e.g., Sales by Region).
- Example: Grouping Sales Data by Region

SQL - GROUP BY



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The GROUP BY clause in SQL is used to group rows with the same values in specified columns and then apply aggregate functions (like COUNT, SUM, AVG, etc.) to each group.

Example:

	StudentID	Name	Age	Email	Department	GPA	GraduationYear
▶	3336	David Palmer	19	sean43@hotmail.com	Mathematics	3.16	2026
	8774	Andrew Roach	23	vbecker@harvey.com	Chemistry	3.75	2027
	1396	Jonathan Gonzalez	22	hollydavis@gmail.com	Physics	2.95	2027
	6716	Kenneth Morrow	24	ganderson@wheeler-atkins.info	Physics	3.55	2029
	8830	Kaitlyn Martinez	18	hayesdiane@gmail.com	Chemistry	2.29	2025
	5305	Tiffany Wolf	23	qanderson@taylor.com	Mathematics	3.3	2029
	5048	James Reyes	20	drodriquez@nauven-cooper.info	Chemistry	2.44	2029

E.g-1: Find the average GPA for each department.



Query:

```
SELECT Department, AVG(GPA) AS Average_GPA
FROM students
GROUP BY Department;
```

Output:

	Department	Average_GPA
▶	Mathematics	3.0806666666666667
	Chemistry	2.9363043478260873
	Physics	3.1239215686274506
	Computer Science	3.0443478260869563
	Biology	2.9155555555555556

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E.g-1: Find the average GPA for each department.

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- Drag "Department" to Rows: This creates groups based on departments.
- Drag "GPA" to Columns: Tableau automatically aggregates the GPA (e.g., SUM, AVG, etc.).
- By default, Tableau applies SUM, but you can change this to AVG by right-clicking on GPA → Aggregate → Average.

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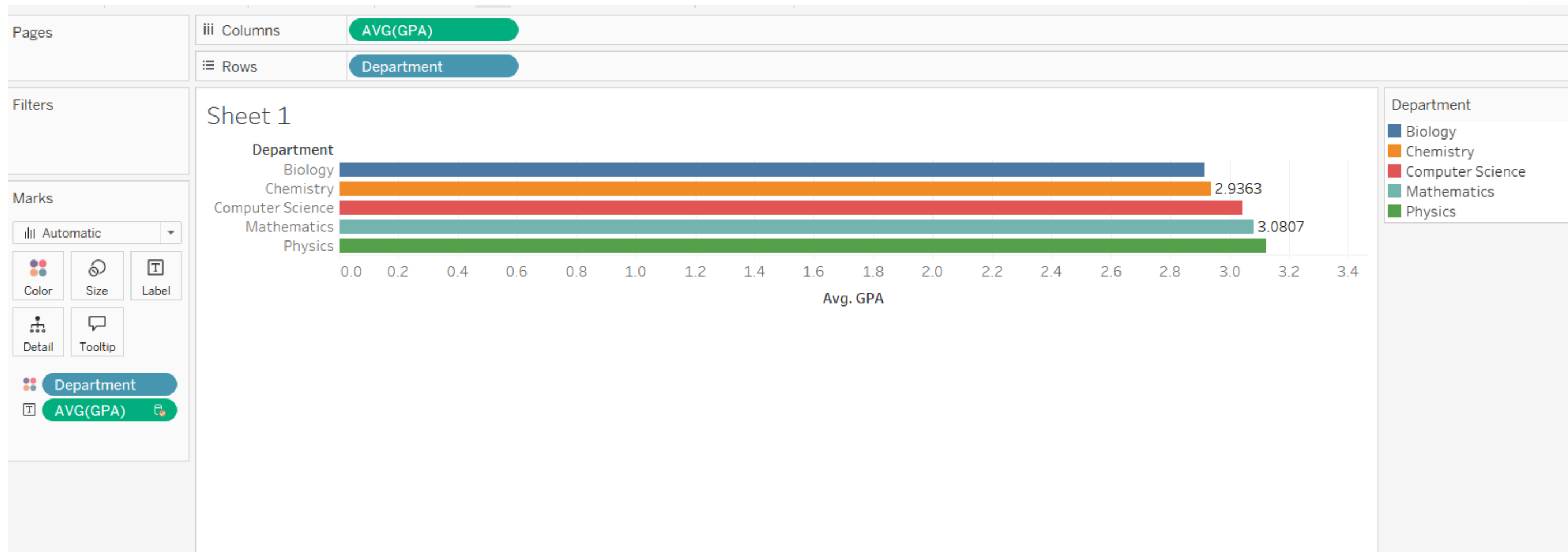
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E.g-2: Count of Students by Department who are above 18



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Query:

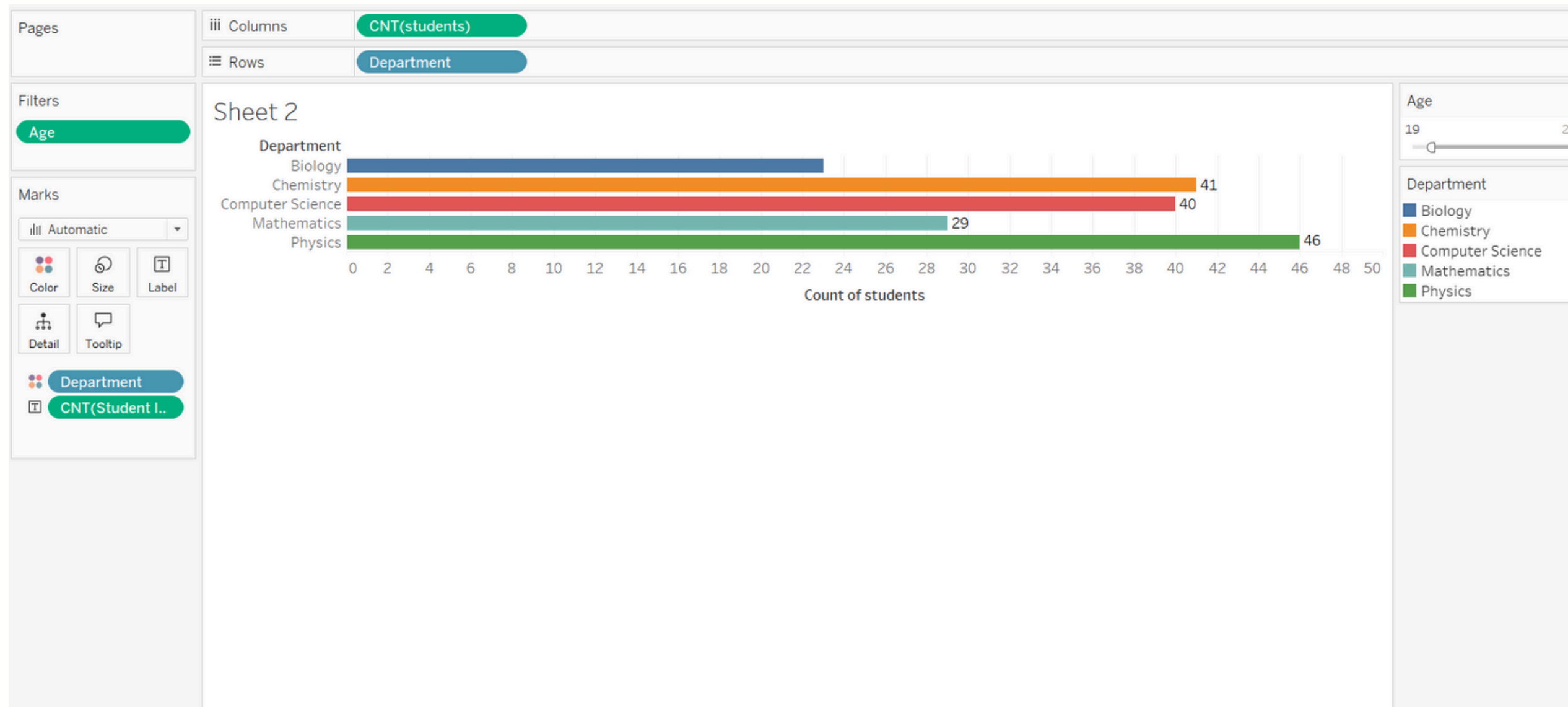
```
SELECT Department, COUNT(StudentID) AS Student_Count
FROM students
WHERE Age > 18
GROUP BY Department;
```

Output:

	Department	Student_Count
▶	Mathematics	29
	Chemistry	41
	Physics	46
	Computer Science	40
	Biology	23

E.g-2: Count of Students by Department who are above 18

- Drag Department to Rows.
- Drag Student ID to Columns.
- Set aggregation to Count:
- Right-click on "Student ID" → Aggregate → Count.
- Drag Age to the Filters shelf, set it to "At least 19," and click OK



E.g-3: Average Age by Graduation Year for the years 2026 - 2028

Query:

```
SELECT GraduationYear, AVG(Age) AS Average_Age
FROM students
WHERE GraduationYear BETWEEN 2026 AND 2028
GROUP BY GraduationYear;
```

Output:

	GraduationYear	Average_Age
▶	2026	21.6957
	2027	21.8333
	2028	21.2800

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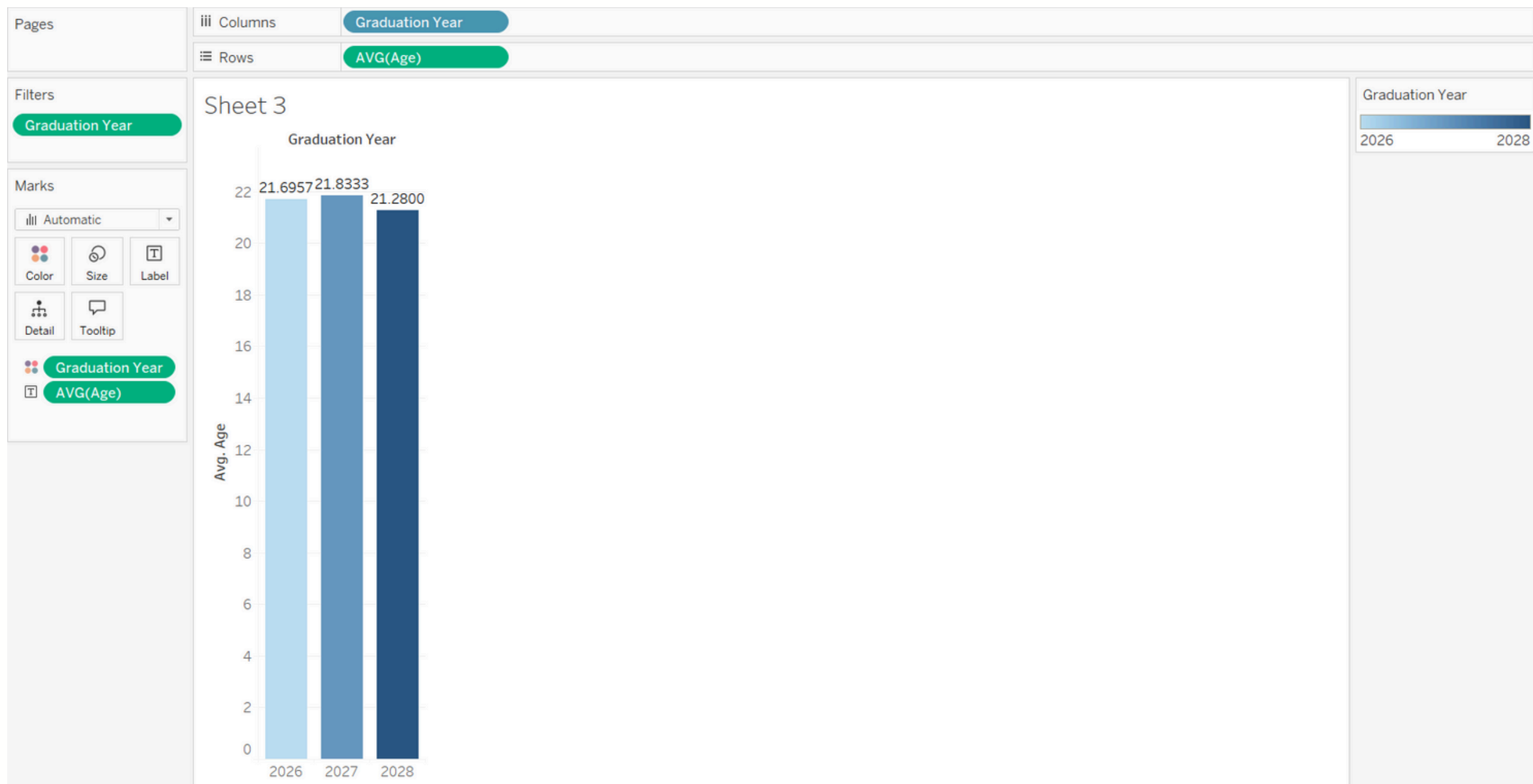
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E.g-3: Average Age by Graduation Year for the years 2026 - 2028

- Drag Graduation Year to Rows.
- Drag Age to Columns.
- Change the aggregation to Average:
- Right-click on "Age" → Aggregate → Average.
- Drag Graduation Year to the Filters shelf, set the range to "2026 to 2028" and click OK



E.g-4: Total number of students for each department and year

Query:

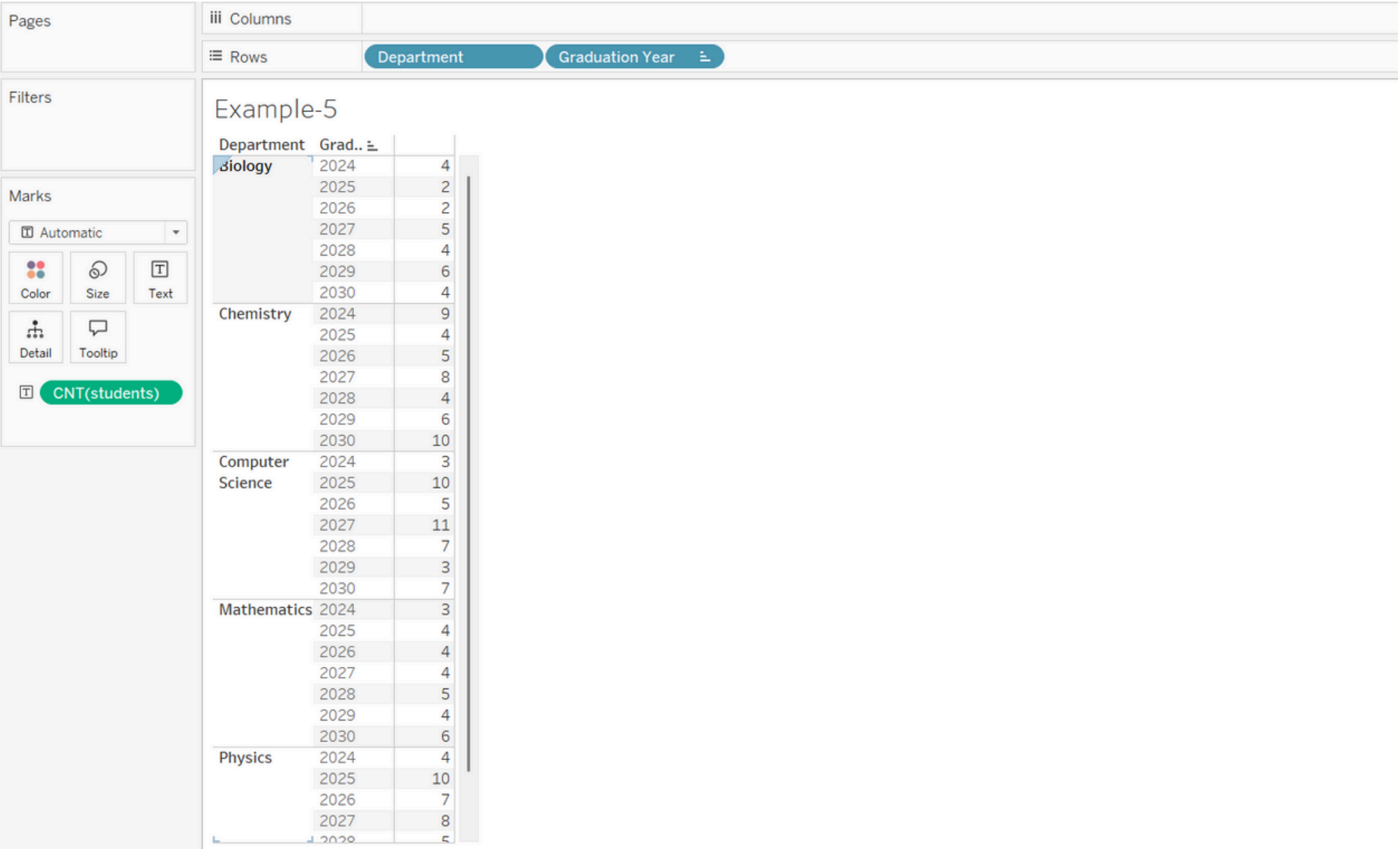
```
SELECT GraduationYear, Department, COUNT(StudentID) AS CountOfStudents
FROM students
GROUP BY GraduationYear, Department
ORDER BY GraduationYear, Department;
```

Output:

	GraduationYear	Department	CountOfStudents
▶	2024	Biology	4
	2024	Chemistry	9
	2024	Computer Science	3
	2024	Mathematics	3
	2024	Physics	4
	2025	Biology	2
	2025	Chemistry	4
	2025	Computer Science	10
	2025	Mathematics	4
	2025	Physics	10
	2026	Biology	2
	2026	Chemistry	5
	2026	Computer Science	5
	2026	Mathematics	4

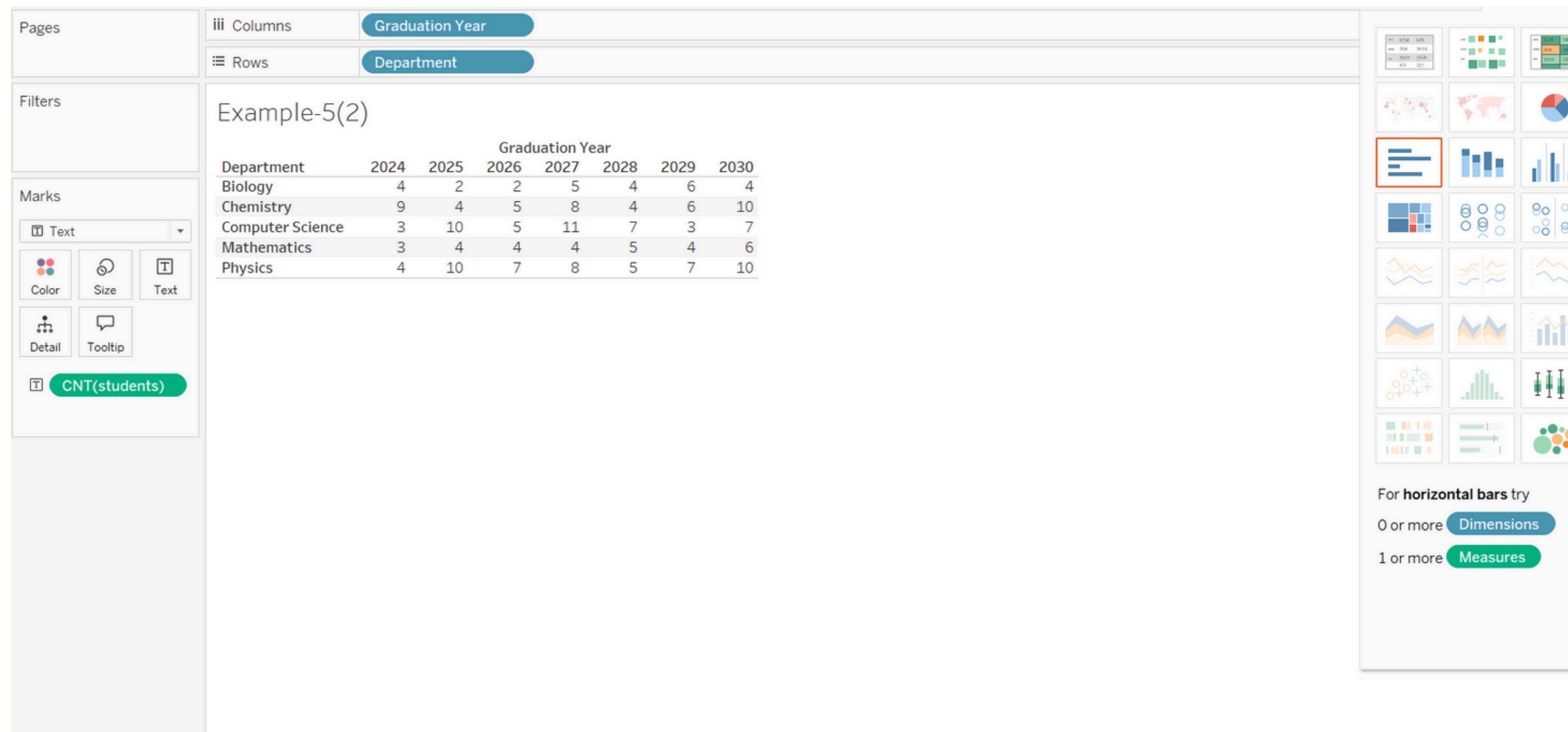
E.g-4: Total number of students for each department and year

- Drag Department to Rows.
- Drag Graduation to Rows again beside Department.
- Drag Count of Students to the Text Label in the Marks Field.



E.g-4: Total number of students for each department and year

- Drag Department to Rows.
- Drag Graduation Year to Columns.
- Change the Type in Marks Field from 'Automatic' to 'Text'.
- Drag Count of Students to the Text Label in Marks.



E.g-5: Total number of students and the average GPA for each department and year

Query:

```
SELECT Department, GraduationYear, COUNT(StudentID) AS student_count, AVG(GPA) AS average_gpa
FROM students
GROUP BY Department, GraduationYear
ORDER BY Department, GraduationYear;
```

Output:

	Department	GraduationYear	student_count	average_gpa
►	Biology	2024	4	2.8249999999999997
	Biology	2025	2	3.8049999999999997
	Biology	2026	2	2.5250000000000004
	Biology	2027	5	3.16
	Biology	2028	4	2.84
	Biology	2029	6	2.6016666666666666
	Biology	2030	4	2.9974999999999996
	Chemistry	2024	9	3.0755555555555554
	Chemistry	2025	4	2.7425
	Chemistry	2026	5	2.5
	Chemistry	2027	8	3.19875

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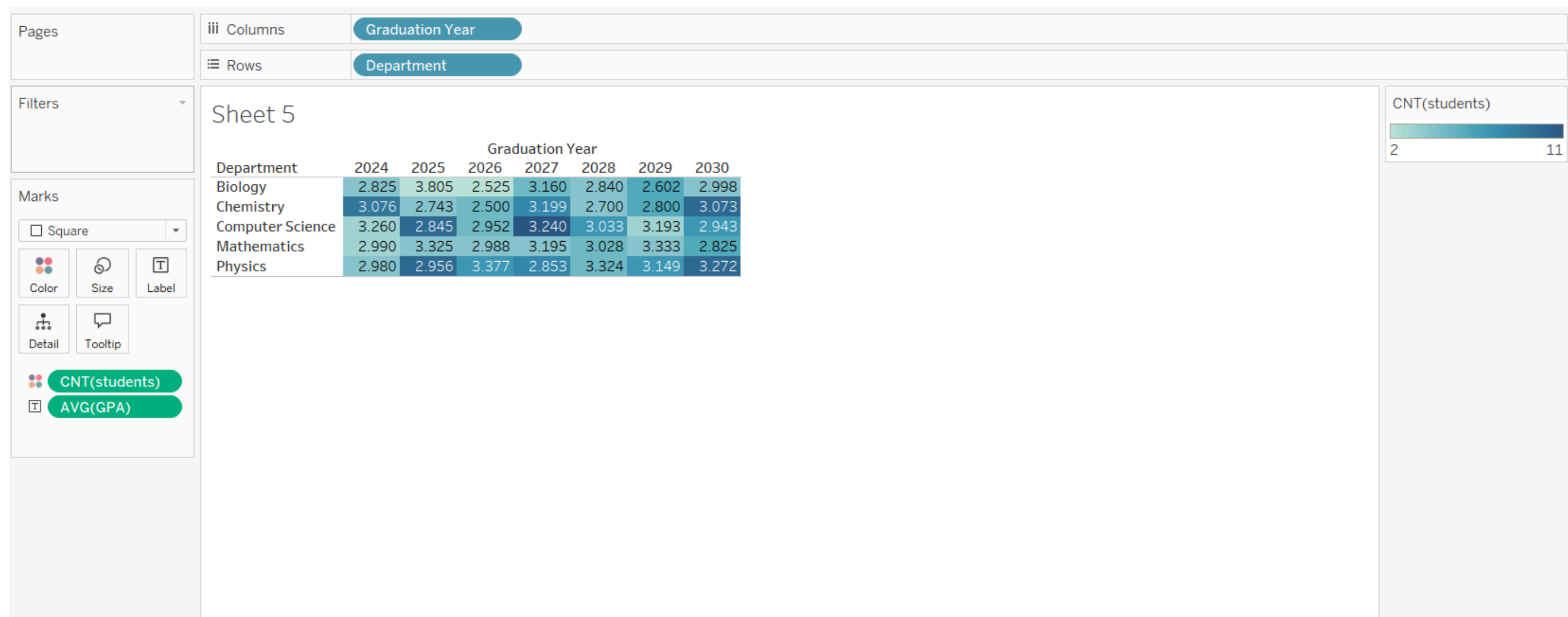
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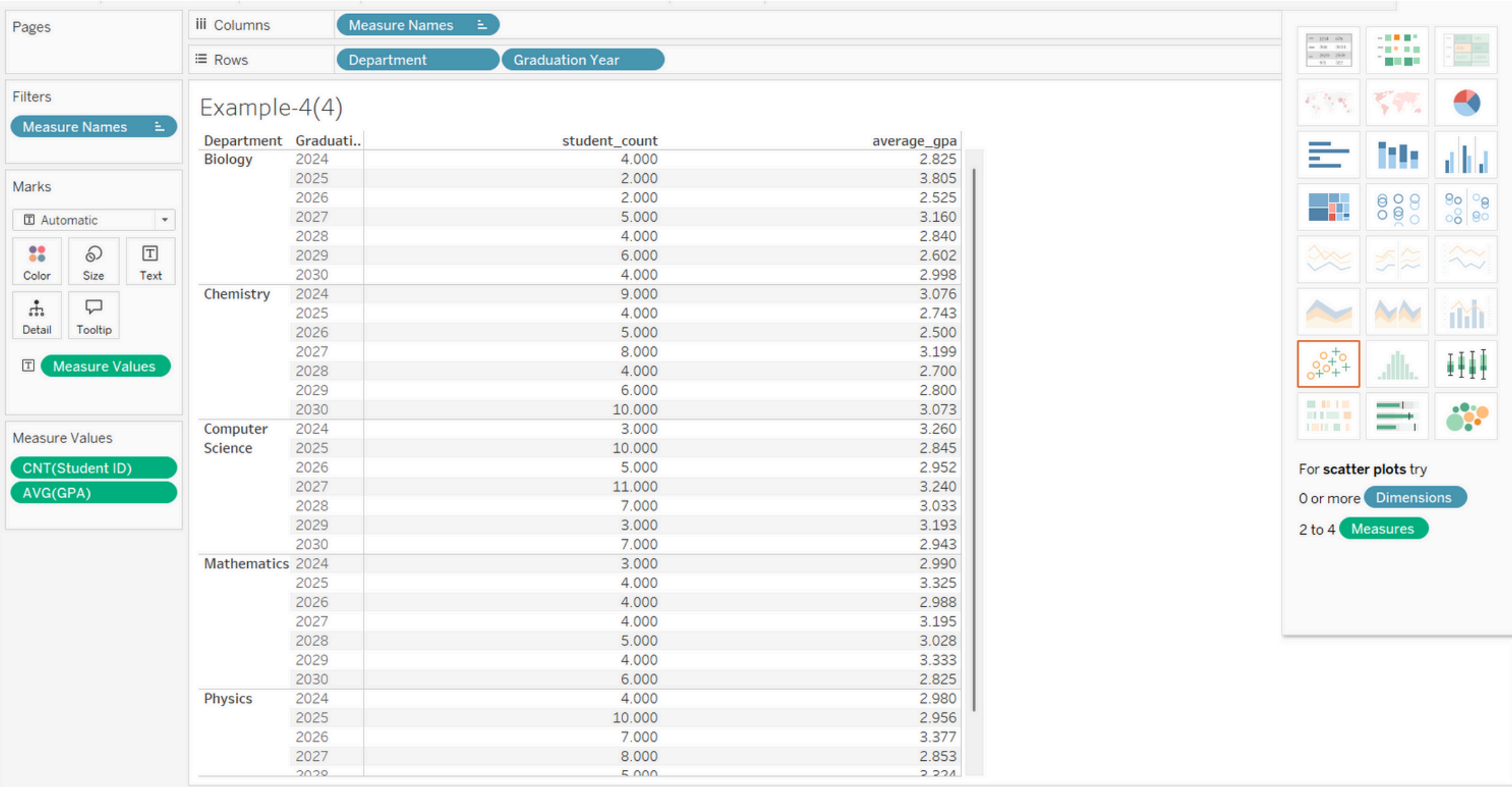
E.g-5: Total number of students and the average GPA for each department and year

- Drag Department to Rows.
- Drag Graduation Year to Columns.
- Drag StudentID to color and change the measure to CNT (count).
- Drag GPA to Label under Marks and change the measure from SUM to AVG.



E.g-5: Total number of students and the average GPA for each department and year

- Drag Department to Rows.
- Drag GraduationYear to Rows, placing it next to Department.
- Drag student_count to Columns.
- Drag average_gpa to Columns, placing it next to student_count.
- Open the Show Me panel (top-right).
- Click the Text Table icon to display the data as a table.





THANK YOU

